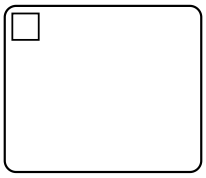
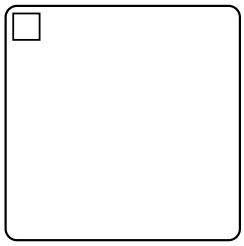
Pip parts tray · 1 of 2 · the parts that go into the device, boxes 4 mm larger than the part

Lay each part on its box as you find it; tick the corner. Print at Actual size, Letter, black.

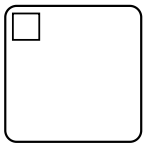
The five modules



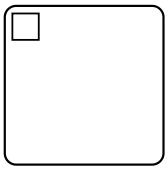
A1 · ESP32-C3 SuperMini
USB at the front



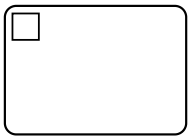
Screen · SSD1306 0.96"
SDA·SCL·VCC·GND



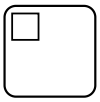
Mic · INMP441
round, hole up



Amp · MAX98357A
pins to the channel

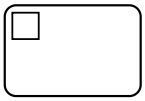


Speaker · CES-20134
mesh up

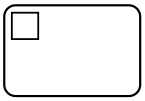


Button
legs left / right

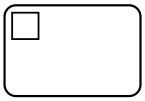
Small parts



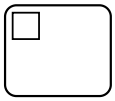
R1 · 4.7 kΩ (or 1 kΩ)
measured: ____



R3 · 10 kΩ
measured: ____



C1 · 220 µF
stripe = −

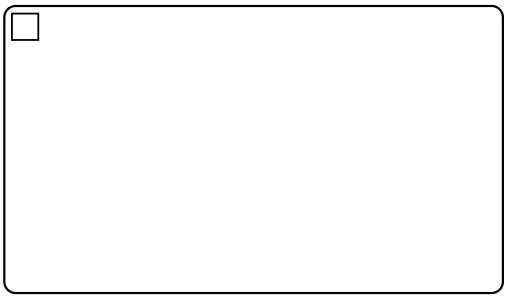


C2 · 100 nF

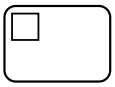
Power



PowerBoost 1000C (copy)
micro-USB · verify pins



Cell · 1200 mAh #258
JST lead, protected



Slide switch

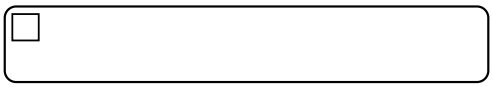


26 AWG red + black
2 × 80 mm

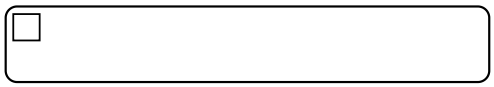


30 AWG
2 × 60 mm

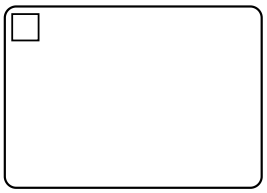
Stock



1.0 mm rod · K&S #9861
or 20 ga nickel silver (0.81)



1.5 mm tube · K&S #9831
or 14 ga nickel silver (frame)



Cardstock
jig card, shelf, cradle

Pip parts tray · 2 of 2 · consumables and tools, boxes not to scale

Lay each part on its box as you find it; tick the corner. Print at Actual size, Letter, black.

Consumables



Kapton tape



Heat-shrink, thinnest



63/37 solder



Chip Quik flux pen



Solder wick



400 grit



91 % IPA



Hot glue sticks

Tools



Soldering station



Electronics tip



Flush cutters



Chain-nose pliers



Round-nose pliers



Calipers



Multimeter



Heat gun



Helping hands



Tweezers



Silicone mat



Hot glue gun



Printed floor template



White sheet to work over